

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

```
#en
2#config t
nter configuration commands, one per line. End with CNTL/Z.
2(config)#hostname R2
2(config)#
2(config)#int f0/0
2(config-if)#ip add 192.168.12.2 255.255.255.0
2(config-if)#no sh
2(config-if)#
2(config-if)#int f0/1
2(config-if)#ip add 192.168.23.1 255.255.255.0
2(config-if)#no sh
2(config-if)#
2(config-if)#router ospf 1
2(config-router)#int f0/0
2(config-if)#ip ospf 1 area 0
2(config-if)#int f0/1
2(config-if)#ip ospf 1 area 0
2(config-if)#en
Ambiguous command: "en"
2(config)#config t
Invalid hex value
2(config)#hostname R2
2(config)#
2(config)#int f0/0
2(config-if)#ip add 192.168.12.2 255.255.255.0
2(config-if)#no sh
2(config-if)#
2(config-if)#int f0/1

R3(config-if)#no sh

R3(config-if)#
R3(config-if)#int f0/1
R3(config-if)#ip add 192.168.34.1 255.255.255.0
R3(config-if)#no sh

R3(config-if)#
R3(config-if)#int f1/0
R3(config-if)#ip add 192.168.2.1 255.255.255.0
R3(config-if)#no sh

R3(config-if)#
R3(config-if)#router ospf 1
R3(config-router)#int f0/0
R3(config-if)#ip ospf 1 area 0
R3(config-if)#int f0/1
R3(config-if)#ip ospf 1 area 0
R3(config-if)#int f1/0
R3(config-if)#ip ospf 1 area 0
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

R3(config-if)#
01:25:17: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.23.1 on FastEthernet0/0 from LOADING to FULL, Loading Done

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
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Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R4
R4(config)#
R4(config)#int f0/0
R4(config-if)#ip add 192.168.34.2 255.255.255.0
R4(config-if)#no sh

R4(config-if)#
R4(config-if)#int f0/1
R4(config-if)#ip add 192.168.14.2 255.255.255.0
R4(config-if)#no sh

R4(config-if)#
R4(config-if)#
R4(config-if)#router ospf 1
R4(config-router)#int f0/0
R4(config-if)#ip ospf 1 area 0
R4(config-if)#int f0/1
R4(config-if)#ip ospf 1 area 0
R4(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

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R1>
R1>en
R1#show ip route
Codes: L - local, C - connected, S - static, E - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, FastEthernet1/0
L    192.168.1.1/32 is directly connected, FastEthernet1/0
O    192.168.2.0/24 [110/3] via 192.168.12.2, 00:06:50, FastEthernet0/0
     [110/3] via 192.168.14.2, 00:06:50, FastEthernet0/1
192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.12.0/24 is directly connected, FastEthernet0/0
L    192.168.12.1/32 is directly connected, FastEthernet0/0
192.168.14.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.14.0/24 is directly connected, FastEthernet0/1
L    192.168.14.1/32 is directly connected, FastEthernet0/1
O    192.168.23.0/24 [110/2] via 192.168.12.2, 00:08:33, FastEthernet0/0
--More--

```

